EC-200 Data Structures

LAB MANUAL # 05

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Student Name: _____

Degree/ Syndicate: _____

	Trait	Obtained Marks	Maximum Marks
R1	**Application Functionality 20%**		20
R2	**Specification & Data structure implementation 30%**		30
R3	**Reusability 10%**		10
R4	**Input Validation 10%**		10
R5	**Efficiency 20%**		20
R6	**Delivery 10%**		10
R7	**Plagiarism above 80%**		1
Total			10

Total Marks = $Obtained\ Marks\ (\sum_1^6 R_i * R_7)$

Lab Manual # 05

Circular Linked List Implementation

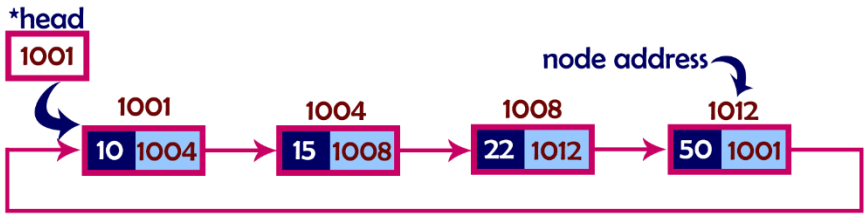
Lab Objective:

To implement Circular Linked List ADT functions.

Lab Description:

In single linked list, every node points to its next node in the sequence and the last node points NULL. However, a **circular linked list is a sequence of elements in which every element has a link to its next element in the sequence and the last element has a link to the first element.** That means circular linked list is similar to the single linked list except that the last node points to the first node in the list

Circular Linked List:



In a circular linked list, we perform the following operations...

1. Insertion
2. Deletion
3. Display

Inserting At Beginning of the list

We can use the following steps to insert a new node at beginning of the circular linked list...

1. Create a **newNode** with given value.
2. Check whether list is **Empty** (**head** == **NULL**)
3. If it is **Empty** then, set **head** = **newNode** and **newNode→next** = **head** .
4. If it is **Not Empty** then, define a Node pointer '**temp**' and initialize with '**head**'.
5. Keep moving the '**temp**' to its next node until it reaches to the last node (until '**temp** → **next** == **head**').
6. Set '**newNode** → **next** =**head**', '**head** = **newNode**' and '**temp** → **next** = **head**'.

Inserting At End of the list

We can use the following steps to insert a new node at end of the circular linked list...

1. Create a **newNode** with given value.
2. Check whether list is **Empty** (**head** == **NULL**).
3. If it is **Empty** then, set **head** = **newNode** and **newNode** → **next** = **head**.
4. If it is **Not Empty** then, define a node pointer **temp** and initialize with **head**.
5. Keep moving the **temp** to its next node until it reaches to the last node in the list (until **temp** → **next** == **head**).
6. Set **temp** → **next** = **newNode** and **newNode** → **next** = **head**.

Inserting At Specific location in the list (After a Node)

We can use the following steps to insert a new node after a node in the circular linked list...

1. Create a **newNode** with given value.
2. Check whether list is **Empty** (**head == NULL**)
3. If it is **Empty** then, set **head = newNode** and **newNode** → **next = head**.
4. If it is **Not Empty** then, define a node pointer **temp** and initialize with **head**.
5. Keep moving the **temp** to its next node until it reaches to the node after which we want to insert the newNode (until **temp1** → **data** is equal to **location**, here location is the node value after which we want to insert the newNode).
6. Every time check whether **temp** is reached to the last node or not. If it is reached to last node then display **'Given node is not found in the list!!! Insertion not possible!!!'** and terminate the function. Otherwise move the **temp** to next node.
7. If **temp** is reached to the exact node after which we want to insert the newNode then check whether it is last node (temp → next == head).
8. If **temp** is last node, then set **temp** → **next = newNode** and **newNode** → **next = head**.
9. If **temp** is not last node, then set **newNode** → **next = temp** → **next** and **temp** → **next = newNode**.

Deleting from Beginning of the list

We can use the following steps to delete a node from beginning of the circular linked list...

1. Check whether list is **Empty** (**head** == **NULL**)
2. If it is **Empty** then, display **'List is Empty!!! Deletion is not possible'** and terminate the function.
3. If it is **Not Empty** then, define two Node pointers **'temp1'** and **'temp2'** and initialize both **'temp1'** and **'temp2'** with **head**.
4. Check whether list is having only one node (**temp1** → **next** == **head**)

5. If it is **TRUE** then set **head** = **NULL** and delete **temp1** (Setting **Empty** list conditions)
6. If it is **FALSE** move the **temp1** until it reaches to the last node. (until **temp1** → **next** == **head**)
7. Then set **head** = **temp2** → **next**, **temp1** → **next** = **head** and delete **temp2**.

Deleting from End of the list

We can use the following steps to delete a node from end of the circular linked list...

1. Check whether list is **Empty** (**head** == **NULL**)
2. If it is **Empty** then, display **'List is Empty!!! Deletion is not possible'** and terminate the function.
3. If it is **Not Empty** then, define two Node pointers **'temp1'** and **'temp2'** and initialize **'temp1'** with **head**.
4. Check whether list has only one Node (**temp1** → **next** == **head**)
5. If it is **TRUE**. Then, set **head** = **NULL** and delete **temp1**. And terminate from the function. (Setting **Empty** list condition)
6. If it is **FALSE**. Then, set **'temp2 = temp1 '** and move **temp1** to its next node. Repeat the same until **temp1** reaches to the last node in the list. (until **temp1** → **next** == **head**)
7. Set **temp2** → **next** = **head** and delete **temp1**.

Deleting a Specific Node from the list

We can use the following steps to delete a specific node from the circular linked list...

1. Check whether list is **Empty** (**head** == **NULL**)
2. If it is **Empty** then, display **'List is Empty!!! Deletion is not possible'** and terminate the function.
3. If it is **Not Empty** then, define two Node pointers **'temp1'** and **'temp2'** and initialize **'temp1'** with **head**.
4. Keep moving the **temp1** until it reaches to the exact node to be deleted or to the last node. And every time set **'temp2 = temp1'** before moving the **'temp1'** to its next node.
5. If it is reached to the last node then display **'Given node not found in the list! Deletion not possible!!!'**. And terminate the function.
6. If it is reached to the exact node which we want to delete, then check whether list is having only one node (**temp1** → **next** == **head**)
7. If list has only one node and that is the node to be deleted then set **head** = **NULL** and delete **temp1** (**free(temp1)**).
8. If list contains multiple nodes, then check whether **temp1** is the first node in the list (**temp1** == **head**).

9. If **temp1** is the first node then set **temp2** = **head** and keep moving **temp2** to its next node until **temp2** reaches to the last node. Then set **head = head → next**, **temp2 → next** = **head** and delete **temp1**.
10. If **temp1** is not first node then check whether it is last node in the list (**temp1 → next == head**).
11. If **temp1** is last node then set **temp2 → next** = **head** and delete **temp1** (**free(temp1)**).
12. If **temp1** is not first node and not last node then set **temp2 → next** = **temp1 → next** and delete **temp1** (**free(temp1)**).

Displaying a circular Linked List

We can use the following steps to display the elements of a circular linked list...

1. Check whether list is **Empty** (**head** == **NULL**)
2. If it is **Empty**, then display **'List is Empty!!!'** and terminate the function.
3. If it is **Not Empty** then, define a Node pointer **'temp'** and initialize with **head**.
4. Keep displaying **temp → data** with an arrow (**--->**) until **temp** reaches to the last node
5. Finally display **temp → data** with arrow pointing to **head → data**.

LAB TASK:

1. **Create templated Circular Linked List ADT and provide**
    a. Constructor / copy constructor
    b. Destructor
    c. InsertAtStart(intval)
    d. InsertAtAnyposition(intposition,intval)
    e. insertAtEnd(intval)
    f. deleteAtstart()
    g. DeleteAtAnyposition(node)
    h. deleteAtEnd()
    i. Traverse()
    j. isEmpty()

Note: Your program should have no memory leakage & dangling pointers. Your program should be validated properly and display warnings and errors to user as well. Reuse functions if needed to increase program efficiency.

TEST PLAN:

1. **Execute your test plan. If you discover mistakes in your implementation of the Circular List ADT, correct them and execute your test plan again.**

Sr.	Operations	Expected Results	Results/Status
1.	Create an integer type list LinkedList<int> **myList** with default constructor	head=NULL	
2.	Copy constructor	Head = NULL	
3.	Check if the list isEmpty?	yes	
4.	Insert values 1-5 in list		
5.	Check if the list isEmpty?	No	
6.	Traverse the list	1,2,3,4,5	
7.	Delete from start		
8.	Traverse the list	2,3,4,5	
9.	Insert at Start (6)		
10.	Traverse the list	6,2,3,4,5	
11.	Insert at End (9)		
12.	Traverse the list	6,2,3,4,5,9	
13.	Delete from end		
14.	Traverse the list	6,2,3,4,5	
15.	Delete from position 3		
16.	Traverse the list	6,2,4,5	
17.	Insert value 7 at position 4		
18.	Traverse the list	6,2,4,7,5	
19.	Create an integer type list LinkedList<float> **myList2** with default constructor	head = NULL	
20.	Insert 5 values.	1.5, 2.5, 3.5, 4.5, 5.5	
21.	Delete from position 2		
22.	Traverse the list	1.5,3.5,4.5,5.5	
23.	Insert at start (5.5)		
24.	Insert at end (6.5)		
25.	Insert at position (5,7.7)		
26.	Traverse the list	5.5,1.5,3.5,4.5,7.7,5.5	

THINK:

1. If you do not provide copy constructor in the above ADT, program will generate a run time error at the time of destructor calling. Why?

2. Consider a small circular linked list. How to detect the presence of cycles in this list effectively?

3. What will happen if you do not provide destructor to your Circular linked list ADT?

4. In which sequence the destructor of circular linked lists will be called? Which list will be destroyed first?